

DATA CLEANING LOG — Airline Reviews Dataset

Column	Issue Identified	Cleaning / Transformation Action	Result / Impact
Unnamed: 0	Auto-generated index column from CSV export	Dropped with df.drop(columns=["Unnamed: 0"])	Cleaner dataset; no redundant index column
Column Names	Mixed case, spaces in column headers (e.g. "Seat Comfort", "Type Of Traveller")	Standardised with .str.lower().str.strip().str.replace(" ", "_") — all columns snake_case	Consistent naming; easy programmatic access to all columns
wifi_&_connectivity	Dropped — likely too sparse or not relevant to core analysis	Removed with df.drop(columns=["wifi_&_connectivity", "aircraft"])	Reduced dimensionality; focus on core rating metrics
aircraft	Dropped alongside wifi column	Removed with df.drop(columns=[...])	Simpler schema; not needed for rating-based analysis
type_of_traveller	Missing values present	Nulls filled with string "Unknown" — preserves all rows for groupby analysis	0 nulls; full data retained for traveller segmentation
seat_type	Missing values present	Nulls filled with "Unknown"	Enables seat-type groupby without row loss
route	Missing values present	Nulls filled with "Unknown"	Route column retained for route-level analysis
date_flown	Missing values present; also stored as string	Nulls filled with "Unknown"; then converted with pd.to_datetime(errors="coerce")	Datetime-ready column; invalid strings become NaT gracefully
seat_comfort, cabin_staff_service, food_&_beverages, ground_service, inflight_entertainment, value_for_money	Missing numeric ratings across multiple sub-rating columns	Nulls imputed with column-wise median — robust to skew; preserves distribution	0 nulls in all rating columns; no row dropped; bias-minimised imputation
overall_rating	Stored as object/string; some non-numeric values present	Converted with pd.to_numeric(errors="coerce"); rows with NaN dropped via dropna(subset=["overall_rating"])	Clean numeric column; only valid ratings retained; minimal row loss
review_date	Stored as string; inconsistent date formats	Converted with pd.to_datetime(errors="coerce")	Datetime format; enables time-series analysis; invalid entries become NaT
recommended	String column with "yes"/"no" values; inconsistent casing/whitespace	Normalised with .str.lower().str.strip(); mapped to binary via {"yes":1, "no":0} into new column is_recommended	Binary numeric flag; usable in correlation, groupby, and ML pipelines
overall_rating (rating_category)	No categorical grouping for rating levels	Engineered new column: rating_category — "Low" (<=3), "Medium" (4–6), "High" (7–10)	Enables segment-level analysis; useful for visualisations and dashboards

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Final Output Columns	Full dataset had many columns not needed for reporting	Final dataset scoped to: airline_name, overall_rating, seat_type, type_of_traveller, is_recommended, rating_category	Lean, focused CSV exported to . ./data/processed/final_airline_reviews.csv